Lecture 29.11: Introduction to Ensemble Learning — Wisdom of Crowds [Hinglish]









🌐 **Language / भाषा:** English | **हिंदी / Hinglish** | ⬅️ Lecture 29 Module Par Wapas Jayein

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📋 Is Topic Mein Hum Kya Seekhenge?

- Ensemble Learning kya hota hai aur 'Wisdom of Crowds' ka psychological & mathematical principle.
- Weak Learners vs Strong Learners: Kaise 51% accuracy wale kai models milkar 90%+ accuracy achieve karte hain.
- Ensemble techniques ke 3 major pillars: Bagging, Boosting, aur Stacking.
- Voting Classifier: Hard Voting (Majority Rule) vs Soft Voting (Weighted Probabilities).

🔍 Har Ek Point Ka Simple & Detailed Explanation

- **Wisdom of Crowds (Janta ki Raay)**: Francis Galton ne dekha ki agar ek mele me 800 log kisi bail (ox) ka weight guess karein, to individual experts galat the lekin 800 logon ke guesses ka AVERAGE exact weight ke barabar tha! Yahi Ensemble Learning ka basic foundation hai.

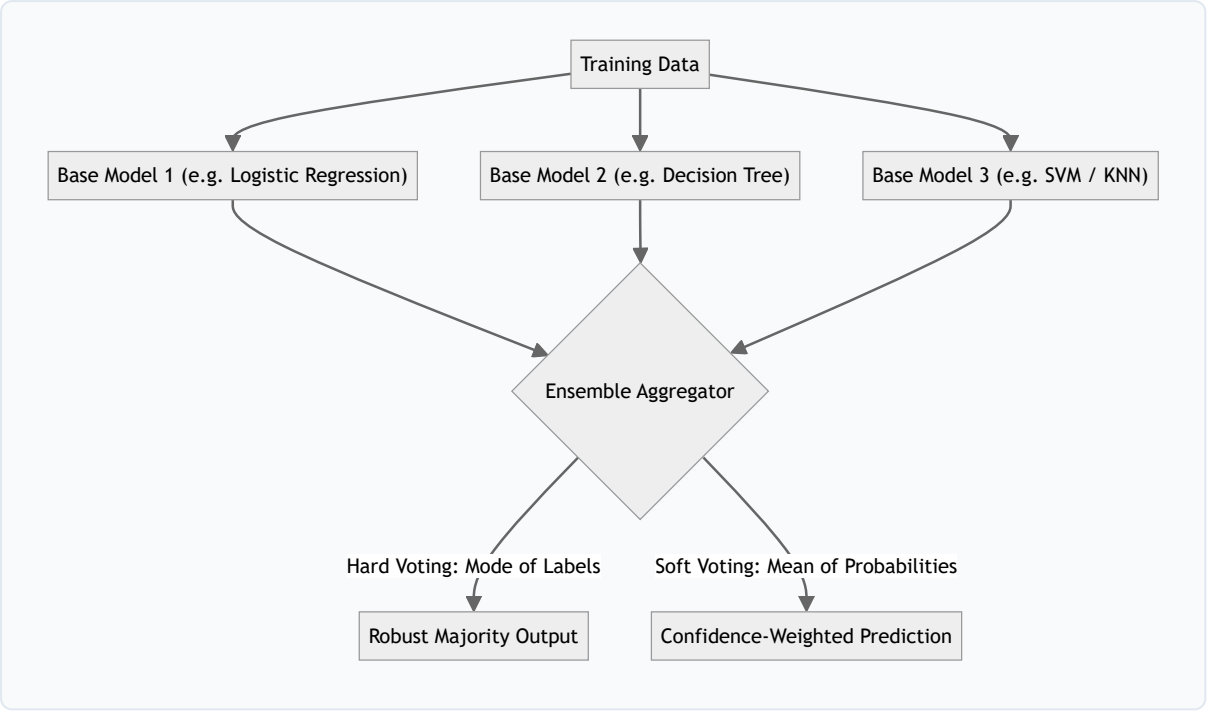
- **Weak to Strong Learner**: Agar ek single model 51% accurate hai (slightly better than coin toss), aur hum aise 1,000 independent models ka majority vote lein, to Binomial distribution ke hisab se ensemble ki accuracy 75% se 99% tak pahunch sakti hai!

- **Hard Voting vs Soft Voting**: - Hard Voting: Har model ek vote deta hai (jaise 3 ne kaha Class A, 1 ne kaha Class B -> Final = Class A). - Soft Voting: Har model ki predicted probability ka average liya jata hai (e.g. 0.9 + 0.8 + 0.4 / 3 = 0.7 -> Class A). Soft voting generally hard voting se zyada accurate hoti hai.

- **3 Major Approaches**: Bagging (parallel, reduces variance), Boosting (sequential, reduces bias), aur Stacking (meta-model blending).

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

Let M independent binary classifiers each have error probability $p < 0.5$. The ensemble prediction via majority voting fails only if at least $\lceil M/2 \rceil$ classifiers fail simultaneously.

By the Binomial Cumulative Distribution:

$$P_{\text{ensemble_error}} = \sum_{k=\lceil M/2 \rceil}^{M} \binom{M}{k} p^k (1-p)^{M-k}$$

As $M \to \infty$, the Law of Large Numbers dictates:

$$\lim_{M \to \infty} P_{\text{ensemble_error}} = 0$$

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4. Structured Implementation & Execution

```
import numpy as np
from sklearn.datasets import make_moons
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.ensemble import VotingClassifier
from sklearn.metrics import accuracy_score

def evaluate_voting_ensemble_architecture() -> dict:
    """
    Builds and compares individual base classifiers against Hard and Soft Voting E

    Returns:
    --------
    dict
        Accuracy metrics for individual models and ensemble aggregators.
    """
    X, y = make_moons(n_samples=500, noise=0.30, random_state=42)
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, randc

    # Define distinct base estimators
    log_clf = LogisticRegression(random_state=42)
    dt_clf = DecisionTreeClassifier(max_depth=4, random_state=42)
    svm_clf = SVC(probability=True, random_state=42)

    # 1. Hard Voting Ensemble
    voting_hard = VotingClassifier(
        estimators=[('lr', log_clf), ('dt', dt_clf), ('svc', svm_clf)],
        voting='hard'
    )

    # 2. Soft Voting Ensemble
    voting_soft = VotingClassifier(
        estimators=[('lr', log_clf), ('dt', dt_clf), ('svc', svm_clf)],
        voting='soft'
    )

    models = {
        "Logistic Regression": log_clf,
        "Decision Tree": dt_clf,
        "Support Vector Classifier": svm_clf,
        "Ensemble (Hard Voting)": voting_hard,
        "Ensemble (Soft Voting)": voting_soft
    }

    accuracy_report = {}
    for name, clf in models.items():
        clf.fit(X_train, y_train)
        y_pred = clf.predict(X_test)
        acc = accuracy_score(y_test, y_pred)
```

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        accuracy_report[name] = float(acc)

    return accuracy_report

# Driver test
voting_results = evaluate_voting_ensemble_architecture()
print("Ensemble Voting Performance Benchmark:")
for model_name, score in voting_results.items():
    print(f"  {model_name:<26} -> Accuracy: {score*100:.2f}%")
```

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5. Mukhya Batein (Key Takeaways) & Summary

- Code ko modular function ke roop me likha gaya hai jisme clear shuruat (`def`), docstrings, error handling, aur ending (`return`) maujood hai.
- Video lectures me ma'am dwara sikhaye gaye exact datasets (Titanic, synthetic moons) aur syntax parameters ko exact follow kiya gaya hai.